

Signal and System Design (IGS2137)

Jump Together, Fly Farther!



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Week 4 Lab

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- Review Script Files
- Nested Loops Breaks
- Activity: Bank Balance
- Activity: Output
- Q & A

Exercise 1: Script Files

Write a script that will prompt the user for a length and units.

Exercise 1: Script Files

Write a script that will prompt the user for a length and units.

- formats printing both length and units.

```
% Prompt the user for a length and units
```

```
length = input('Enter the length: ');  
units = input('Is that (f)eet or m(eters)? : ', 's');  
fprintf('The character is %3c.\n', units)  
fprintf('The number is %-8.3f.\n', length)
```

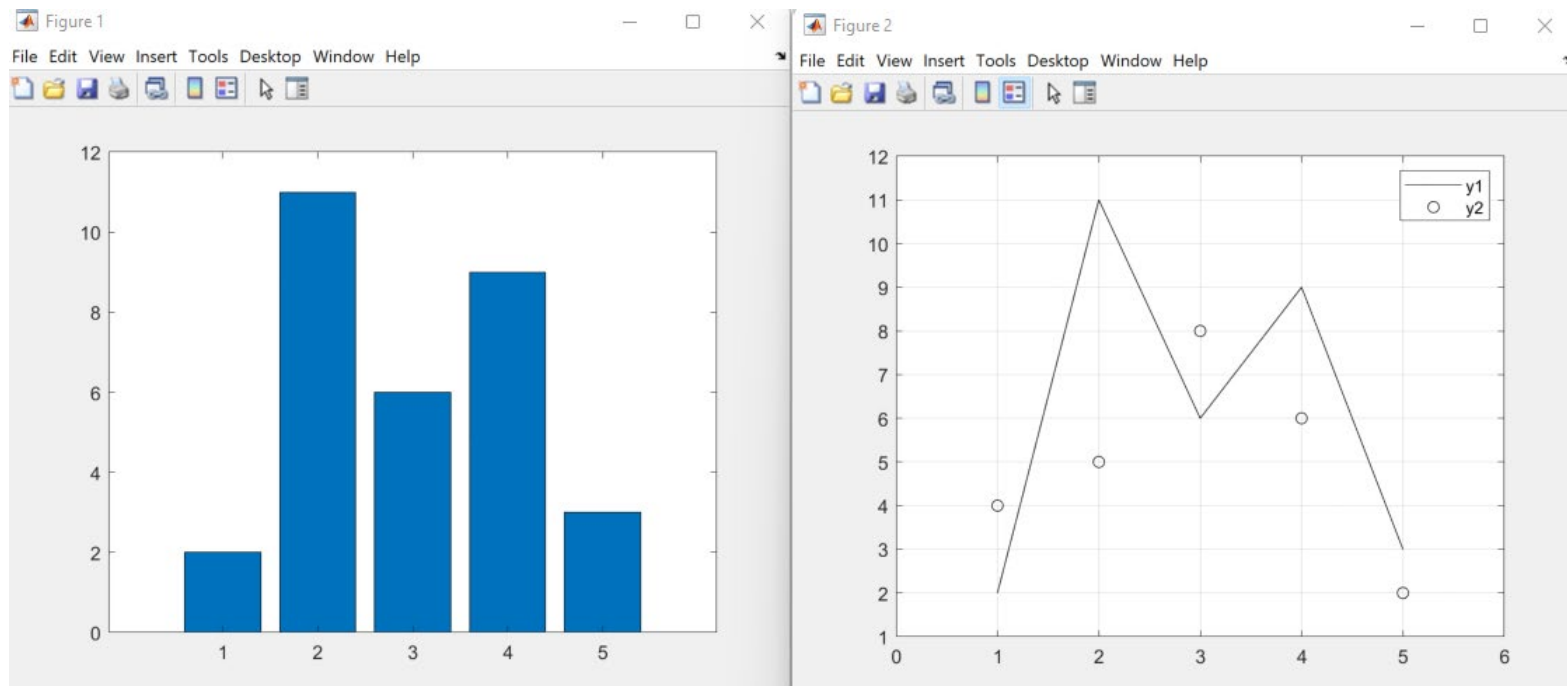
Reference:

https://www.mathworks.com/help/matlab/referencelist.html?type=function&s_tid=CRUX_topnav

Exercise 2: Script Files

Write a script that will plot the following figures.

Note that both plots use the same data for x , $y1$ and $y2$ arrays.



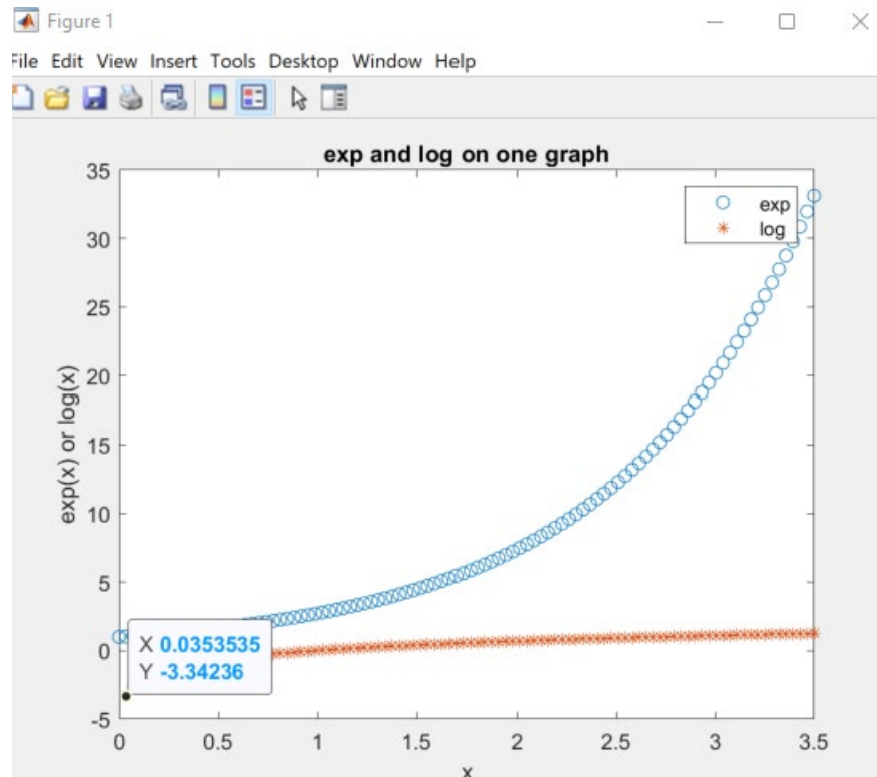
Exercise 2: Solution

% This creates 2 different plots, in 2 different
% Figure Windows, to demonstrate some plot features

```
clf
x = 1:5; % Not necessary
y1 = [2 11 6 9 3];
y2 = [4 5 8 6 2];
% Put a bar chart in Figure 1
figure(1)
bar(x,y1)
% Put plots using different y values on one plot
% with a legend
figure(2)
plot(x,y1,'k')
hold on
plot(x,y2,'ko')
grid on
legend('y1','y2')
axis([0 6 1 12])
```

Exercise 3: Script Files

Write a script that plots $\exp(x)$ and $\log(x)$ on one graph.



Exercise 3: Solution

```
% Plots exp(x) and log(x) on one graph

x = linspace(0,3.5);
ey = exp(x);
ly = log(x);
plot(x,ey,'o')
hold on
plot(x,ly,'*')
legend('exp', 'log')
xlabel('x')
ylabel('exp(x) or log(x)')
title('exp and log on one graph')
```


Exercise 3: Script Files

Write a script that

- prompts the user for the size of a matrix,
- generates a matrix of random integers,
- and save to a file.

Exercise 2: Solution

```
% Prompt the user for size of a matrix, generate  
% a matrix of random integers, and save to a file  
  
r = input('Enter the number of rows: ');  
c = input('Enter the number of columns: ');  
mat = randi(100,r,c);  
save pp2dot7.dat mat -ascii
```

SCRIPT FILES (NESTED LOOPS BREAKS)

Nested-Loop: Simple Example

What will the following code produce?

```
for r = 1:5
    for c = 1:4
        fprintf('r = %i and c = %i \n',r,c);
    end
end
```

Adding Break Statements

What happens when we add a break statement in the outer loop?

```
for r = 1:5
    for c = 1:4
        fprintf('r = %i and c = %i \n',r,c);
    end
    if r == 2
        break;
    end
end
```

Adding Break Statements

What happens when we add a break statement in the inner loop?

```
for r = 1:5
    for c = 1:4
        if r == 2
            break;
        end
        fprintf('r = %i and c = %i \n',r,c);
    end
end
```

Activity: Bank Balance

Suppose you open a bank account with \$1000. The annual interest rate is 5% and interest is compounded monthly. However, you want to determine how much the value of your account will change after 10 years for different monthly deposits (\$0, \$100, \$200, \$300, \$400, \$500).

Write a script that will determine the value in your account after 10 years for each monthly deposit.

Activity: Output

Write a script that will display each of the following shapes using asterisks *

The user should be able to specify the size of each shape

```
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
```

Solid Square

```
* * * * *
*       *
*       *
*       *
*       *
* * * * *
```

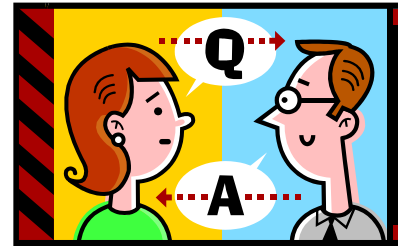
Open Square

```
      *
    * * *
  * * * * *
* * * * * * *
```

Triangle

Brief break (if on schedule)

Q&A?



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